Mandatory Assignment 02

Natural Language Processing and Text Analytics (CDSCO1021C)

Ask question to: Rajani Singh (`rs.digi@cbs.dk`)

Deadline: {check your digital exam for exact date and time `https://eksamen.cbs.dk/`}

Please Read the Instructions Before Proceeding

1. This mandatory assignment consists of two main parts. In Part 1, you have to choose any one topic to answer, while Part 2 is mandatory.
2. Please use Python 3. It is highly appreciated if you use comments extensively in your code.
3. The Python code for answering the questions can be submitted either as one single *Jupyter* notebook or multiple *Jupyter* notebooks. Alternatively, it can be submitted as one or more Python files (with a .py extension).
4. Please note that you need to upload your solutions before the deadline on **Canvas**. Submit *only one zip file* named after your group. For example, if you are 'team 7', then your solutions should be named 'Team-7.zip'. It is important to include your names and student numbers on each solution file.

Part 1: Written Assignment - Select Any Topic

Question 1

Write a brief extended abstract (of approximately 2-pages) reflecting on the application of any **one** (even you are free to combine both approaches if you like) of the following NLP techniques.

1. Lexicon based sentiment Analysis
2. Topic modelling

To answer the above question, you can also do a tiny literature review to find out how your chosen technique(s) is/are used across various domains to address/solve which kinds of problems. Alternatively, you can also write a report on the basic principles of the chosen technique(s) and explain the foundations/fundamentals behind it and how it works, maybe using some easy example to understand it. Therefore you have several choices to write/shape your extended abstract and feel free to choose whatever direction you want to explore.

Your report should conform to general formatting guidelines and academic standards that are expected for written projects at CBS and use a proper referencing/citations style (e.g. APA, MLA, Harvard, etc.) for your report.

Part 2: Mandatory Question

Question 2

There is uploaded dataset named 'cyberbullying-tweets.csv'. Alternatively you can download from here: `https://www.kaggle.com/datasets/soorajtomar/cyberbullying-tweets`
For this dataset, build a Naive Bayes classifier and Logistic Regression classifier for binary classification of the comments as *cyberbullying* or *not cyberbullying*. To complete this question you have to do the following tasks:

1. Data Preparation (use scikit-learn [1] or NLTK [2])
   - Load the dataset using pandas and do basic data exploration (check for data imbalance, read and understand the columns, check for missing values).
   - Preprocess the text data (tokenize, lowercase, remove stop words etc.) and vectorization (choose either Bag-of words or TF-IDF).
   - Split the data into 80% training and 20% testing sets.
2. Naive Bayes Classification (use scikit-learn [1] or NLTK [2])
   - Train a Naive Bayes classifier on the training set. Choose an appropriate variant of Naive Bayes based on your data preprocessing(Bag-of words or TF-IDF).
   - Evaluate the classifier performance on the test set by computing accuracy, precision, recall, and F1-score.
3. Logistic Regression Classification (use scikit-learn [1])
   - Train a Logistic Regression classifier on the same training set.
   - Evaluate the classifier performance on the same test set by computing accuracy, precision, recall, and F1-score.
4. Write one page short report that includes
   - Models' comparison (Naive Bayes model with Logistic Regression model).
   - Write $2-3$ advantage and disadvantage for both models specifically for this dataset.
   - Discuss any other important finding (if any).

References

[1] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[2] S. Bird, "Nltk: the natural language toolkit," in *Proceedings of the COLING/ACL on Interactive presentation sessions*, pp. 69–72, Association for Computational Linguistics, 2006.